

Generation Interconnection Request System Impact Study Report

For Queue Project Q12-0144

Pangborn 138 kV
May 16, 2014

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q12-0144. The Interconnection Customer proposes a 169.5 MW wind facility interconnecting at the Pangborn 138 kV substation in the Maryland area. The request was received on July 20, 2012. The study evaluated system intact and single contingency conditions using the regional planning case, including 97 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 7 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$9,164,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of Bells Mill-Beverly Farms 138.0 kV line, the Tramway-Rossmoor South 138.0 kV loads to 125.7% of its 247 MVA rating. Project DFAX: 14.4%.

The Fanagan-Seek Lane 138.0 kV is overloaded to 121.3% of its 271 MVA emergency rating for the loss of the Darien-Colesville 138.0 kV line. The project contributes 6.7% DFAX (11.3 MW) to this overload.

Polk-Bells Mill 138.0 kV: 118.8% loading (230 MVA) following the outage of the Twinbrook Substation-Parklawn 138.0 kV line; DFAX = 6.3%.

The Rossmoor S-Trumann 138.0 kV L is overloaded to 112.3% of its 229 MVA emergency rating for the loss of the Bells Mill-Beverly Farms 138.0 kV line. The project contributes 14.4% DFAX (24.4 MW) to this overload.

Twinbrook-Bells Mill 138.0 kV: 108.3% loading (220 MVA) following the outage of the Polk-Bells Mill 138.0 kV; DFAX = 22.8%.

For the loss of Tramway-Rossmoor S 138.0 kV L, the Bells Mill-Beverly Farms 138.0 kV loads to 105.8% of its 212 MVA rating. Project DFAX: 31.9%.

The Darien-Trumann 138.0 kV Line is overloaded to 102.3% of its 216 MVA emergency rating for the loss of the Bells Mill Substation-Beverly Farms 138.0 kV line. The project contributes 14.4% DFAX (24.4 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Rossmoor South-Tramway 138.0 kV	Rebuild line with higher-capacity conductor	\$1,224,000	100% (\$1,224,000)
Seek Lane-Flanagan 138.0 kV L	Rebuild line with higher-capacity conductor	\$810,000	100% (\$810,000)
Polk-Bells Mill 138.0 kV	Rebuild line with higher-capacity conductor	\$1,075,000	100% (\$1,075,000)
Rossmoor South-Trumann 138.0 kV	Rebuild line with higher-capacity conductor	\$4,636,000	100% (\$4,636,000)
Twinbrook-Bells Mill 138.0 kV	Reconductor line	\$1,184,000	25% (\$293,000)
Bells Mill-Beverly Farms 138.0 kV	Reconductor line	\$869,000	100% (\$869,000)
Darien-Trumann 138.0 kV Line	Reconductor line	\$257,000	100% (\$257,000)
	Total allocated to project		\$9,164,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.